

STATS 207: Time Series Analysis Project Report

Forecasting NYC Taxi Demand: A Time Series Analysis of Peak Times

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Abstract

Taxi traffic patterns are a critical aspect of urban infrastructure planning. The New York City Taxi and Limousine Commission provides publicly available data about the details of every registered taxi ride that takes place in New York City. Specific cycles and trends can be identified within this data that provide valuable insights into demand for taxis in the city. We developed a number of different models that utilize these patterns and additional data like weather to make forecasts about taxi demand. The models that we studied included but were not limited to linear regression, SARIMA, TBATS, Random Forest regression, and K-Nearest Neighbors regression. We found that a number of different models do a fairly good job at forecasting taxi demand, but K-Nearest Neighbors regression with inputs including the hour, day of week, month, day of month, weather, holidays, and periodic Fourier terms outperformed other models in predicting taxi traffic.

Introduction

Urban planners, commuters, pedestrians, and many others benefit from anticipating taxi traffic patterns ahead of time. There are almost 14,000 registered cabs in New York City[1], so these vehicles make up a substantial portion of the motorized traffic. Accurate demand forecasts allow for better resource allocation, thus reducing congestion and ensuring the availability of taxis when they are most needed. By understanding when and where demand peaks, cities can improve service levels, reduce wait times for passengers, and avoid over-supply in less busy areas. Accurate demand forecasts also can enable taxi companies to manage their fleets effectively, contributing to a more sustainable and cost-efficient transportation system.

Classical time-series techniques along with cutting-edge machine learning approaches can be used to make predictions about taxi demand using historical data. These forecasts could be utilized to make the previously described improvements in urban planning. Numerous studies have shown promise in creating models that can accurately forecast traffic. In this paper, we build upon previous attempts and develop our own novel ideas in an effort to predict hourly yellow taxi pickups across New York City. We analyze large publicly-available datasets to extract patterns and build a number of different predictive models, each with its own strengths and weaknesses. Numerous experiments are designed based upon the nature of our problem and the data in order to ensure scientific rigor. We provide analyses on the performance of the models and insights into the specific architectures that were chosen and why they were selected. A discussion on potential future expansion of our work and any limitations of results is also provided. Overall, we believe that our work provides strong evidence that taxi demand can be accurately forecasted in New York City using the right approaches.

Related Work

There are a number of existing studies that are fairly similar to the work that we have conducted. Researchers have previously attempted to predict taxi demand in New York City using an ARIMA model, but their dataset and exploration of different models were both much more limited than ours[2]. Another study used large-scale datasets to model taxi demand in New York, but they mostly focused on different regression models rather than specific classical time-series approaches[3]. Studies have been done on the use of machine learning (ML) techniques in predicting taxi demand using model inputs such as weather[4], but this work in particular did not focus on New York, nor did it weight the advantages/disadvantages of ML techniques with more classical time-series approaches. Some of the more novel models, like TBATS, that we employed have been consistently used to make forecasts, but not in the realm of taxi demand as far as we know[6]. K-Nearest Neighbors has been used in many instances for time-series forecasting in many fields, but again, not to our knowledge in the realm of taxi demand[7, 8].

Data

The data with which we chose to work is the NYC taxi trip data from the New York City Taxi and Limousine Commission. This data is available on their website[1]. The raw data includes fields capturing pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts for both yellow and green taxis over the time period covering January 2009 through September 2024. It is important to note that within our project we chose to focus solely on the data corresponding to yellow taxis due to the fact that there are limitations on where green taxis are permitted to travel.

With our goal of predicting taxi demand in mind, we decided to drop information regarding trip distances, fares, rate types, and payment types from the data, as it is not particularly relevant. Our dependent variable throughout the project consisted of trip counts aggregated up to the hourly level. It is very important to note that the pickup count exhibits three very strong elements of seasonality over time—there is a daily, weekly, and monthly element, as demonstrated in figures 1 and 2. From the pickup dates and times, we extracted the hour, month, and day of week to use as predictive variables in the models that we constructed.

After doing some research into the effects of weather patterns on taxi demand, we also decided to include historical hourly weather data for New York City that was sourced from Open Meteo’s Free Open-Source Weather API[9]. These weather conditions include information like the temperature, wind, precipitation, and general descriptions like clear, cloudy, etc. We are making what we believe to be the reasonable assumption that in a real-world application of our work, one would have access to fairly accurate weather forecasts ahead of time.

All together, our data consisted of about 24,000 entries of hourly granularity—with each entry containing a pickup count, pickup hour, day of week, month, an indicator for if that day was a holiday or not, and weather conditions as described above.

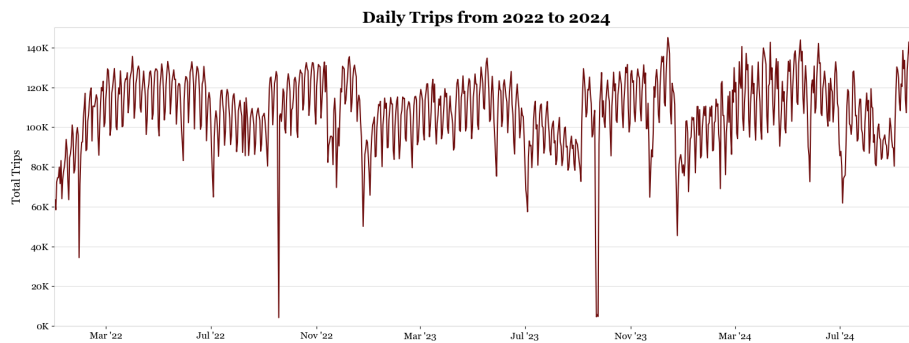
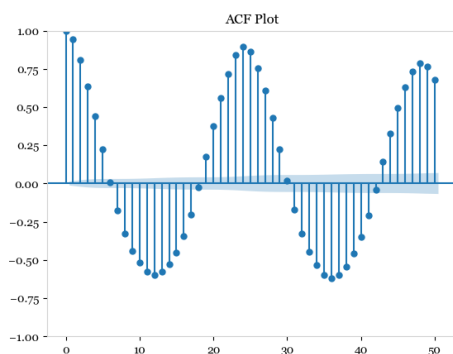
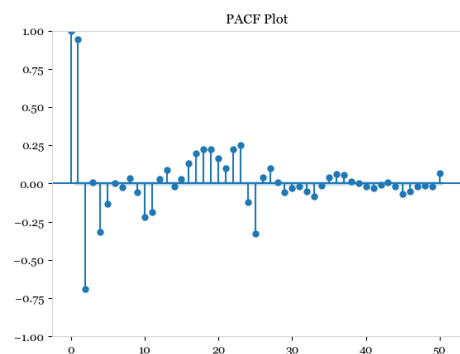


Figure 1: Time plot of hourly trips from Jan 2022 to Sep 2024



(a) ACF



(b) PACF

Figure 2: ACF and PACF plots of Hourly Trip Data

Methods

We developed a number of different models in our efforts to forecast yellow taxi demand. We decided to construct our models from scratch, as there aren't necessarily robust models that have already been developed for our specific needs.

It is important to note that during the training and testing of our models, we focused on the data post-2021. We decided to discard so much data (i.e., that of 2009-2021) due to distributional shocks from the COVID-19 pandemic. During the pandemic, taxi demand plummeted and did not start to recover until about the beginning of 2022. Including this tumultuous data in our train/test sets would negatively impact the development of our models as they would be thrown off by the sudden shifts in distribution. By focusing on 2022 and onward, we are ensuring that our forecasts are particularly relevant and useful in the current day.

We chose to train and test our models in two different ways. First, we utilized a train/test split by training each model on the data ranging from 1/2022 through 8/2024 and using 9/2024 as our out-of-sample test set.

The second manner in which we trained and tested our models utilized time-series cross-validation techniques. We chose to cross-validate in the style of Sklearn's `TimeSeriesSplit` method. This is a variation of k-fold cross-validation which gives the first k folds as the train set and the kth fold as the test set (thus preserving the temporal order of the data). Unlike standard cross-validation methods, successive training sets are supersets of those that come before them. All surplus data is added sequentially to the first training partition, which is always used to train the model.

We chose to train and test in these two distinct manners as a method of ensuring robustness of results. It is important to note that we were not able to cross-validate the SARIMA model nor the TBATS model because of computational limitations. Below, we describe each model in detail. Please note that this is not an exhaustive analysis of all the models we tried, but simply the best performing/most relevant ones that were developed for this study.

Baseline Linear Regression

We constructed a linear regression model as a baseline for our methods. The linear model trains quickly and is easily interpretable. However, the model fails to pick up on elements of seasonality in the data so it is not particularly well-suited to this problem. Within the model, the exogenous variables are hour, day of week, month, and day of month, and the endogenous variable is the passenger pickup count.

SARIMA

The next model that we developed was a Seasonal Autoregressive Integrated Moving Average (SARIMA) model. The biggest drawback of the SARIMA model is that it can only account for one element of seasonality. As previously mentioned in the Data section, the demand for taxis exhibits three different elements of seasonality—daily, weekly, and monthly. We developed many different variations of this model, trying to account for the different elements of seasonality. We also experimented with including exogenous variables like the hour, day of week, month, day of month, weather, and holiday information. Ultimately, we found that the most effective SARIMA model was a $SARIMA(2,0,3) \times (2,1,0,24)$ model with all of the available exogenous variables included. We found these specific parameters by minimizing root mean squared error over a grid search.

TBATS

Due to the drawbacks of SARIMA, we began to research models that we could utilize in order to account for all three elements of seasonality. In our search, we came across the TBATS model. "TBATS" loosely stands for "Trigonometric seasonality, Box-Cox transformation, ARMA errors, Trend, and Seasonal components." The way that TBATS works is it applies a Box-Cox transformation to the data, then models it as a linear combination of an exponentially smoothed trend, seasonal components, and an ARMA component. The seasonal components are modeled by trigonometric functions using Fourier analysis.

It is important to note that TBATS is purely a time-series method. There are no exogenous variables like day of week, weather, etc. included in the model. TBATS performs fairly well due to its ability to account for multiple components of seasonality, but is very much limited by its computational complexity. Just to fit a TBATS model on the roughly 2.5 years of training data took almost three hours.

Random Forest

We also constructed a Random Forest (RF) regression model for our study. Random Forest regressions are relatively computationally efficient and can account for complex temporal relationships in data. We experimented with many variations, but ultimately in our Random Forest regression, we included all of the available exogenous variables including hour, day of week, day of month, month, weather, and holiday information. As described further in Experiments, this method worked particularly well because it struck a great balance between complexity and efficiency.

We also conducted a parameter grid search in order to find an optimal model. We found that the best performance came from a RF regression model with 100 estimators and a max depth of 20. Using 100 estimators seems to strike a good balance between training time and forecasting ability. A depth of 20 seems to be a balanced choice in this case. It's deep enough to capture complex patterns in the data, but not too deep to overfit.

KNN Regression

The final model that we developed was a K-Nearest Neighbors (KNN) regression model. KNN regression models are popular in time-series forecasting due to their ability to forecast based upon recurring events (i.e., clustering demand around certain times of the day, week, etc.). We fit our KNN regression with all available exogenous variables including hour, day of week, day of month, month, weather, and holiday information. We also created Fourier terms for use as input into the model. Fourier terms are essentially periodic functions that capture cyclical patterns of different frequencies in the data. These terms help better model seasonal components of the data.

A grid search was conducted over various parameters including the number of neighbors, weight function, and distance metric for the KNN regression. We found it most optimal to use 7 neighbors with distance weights and the Manhattan distance metric. Utilizing prediction weights that are proportional to the inverse distance between two points means that closer neighbors of a query point will have a greater influence than neighbors which are further away. Thus the past taxi demand for hours of similar circumstances (i.e., similar hour, day of week, etc.) will hold the greatest weight in our predictions. This makes intuitive sense. For example, we would expect that 8am on a rainy Monday in April and 8am on a rainy Monday in May would have similar taxi demand patterns.

Experiments

As previously mentioned, we approached training/testing in two different ways. The single train/test split (training on 1/2022-8/2024 and testing on 9/2024) is advantageous in a sense, but also poses serious limitations. This method of training/testing is good in the sense that it is computationally efficient and allows an immediate look into how our model would have performed if we would've used it in a very recent time period (i.e., last month). Two of our models (TBATS and SARIMA) are very computationally expensive to fit. TBATS takes hours to fit a single model to a single dataset, so cross-validating many times over different datasets would be infeasible unless we utilized some sort of high performance computing cluster. That being said, training/testing over a specific time frame carries the risk of the results being heavily dependent on the specific train/test split and does not guarantee that a model will generalize well.

For performance metrics, we have chosen to include the Root Mean Squared Error (RMSE), the Mean Absolute Error (MAE), the Mean Average Percentage Error (MAPE), and the accuracy as they are common and interpretable metrics across both traditional time series and machine learning methods. We have not included the Akaike information criterion (AIC) as we have included machine learning models like Random Forest that do not have an easily calculated/interpretable AIC. We summarize the metrics below:

$$\begin{aligned}\text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \\ \text{MAE} &= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\ \text{MAPE} &= \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \\ \text{Accuracy} &= 1 - \text{MAPE}\end{aligned}$$

Where n is the number of observations, y_i is the actual value of the i -th observation, $\hat{y}_i$ is the predicted value of the i -th observation.

Table 1 shows the results of training and testing the various models that we have introduced over the single train/test split.

Model	RMSE	MAE	MAPE	Accuracy
Linear Regression	2007.47	1716.09	93.55%	6.45%
SARIMA	2060.26	1698.39	68.74%	31.26%
TBATS	1370.88	1056.94	36.35%	63.65%
Random Forest	1251.71	807.56	14.78%	85.22%
KNN Regression	1041.37	689.07	13.46%	86.54%

Table 1: Performance Metrics of Single Train/Test Split

While SARIMA and TBATS are certainly improvements over the baseline linear model, they fail to perform as well as Random Forest regression and KNN regression. This is likely due to the fact that SARIMA can only account for one element of seasonality and TBATS does not account for any exogenous variables. Random Forest and KNN regression incorporate predictor variables and are able to capture complex temporal relationships. Overall, the KNN regression outperforms all of the other models (even the more computationally expensive ones). As previously described, the KNN regression combines many factors like time patterns and Fourier terms to produce relatively accurate estimates in a computationally feasible manner, making it an ideal model for our forecasting goals.

The second way in which we trained and tested was using a time-series cross-validation technique. We chose to use three folds for our cross-validation. This parameter was chosen because it gives enough lookback for the models to effectively train in the beginning, but still allows us to evaluate over multiple time horizons. In our cross-validation analysis, we did not include the SARIMA nor the TBATS models. Fitting six of these models would require about 18 hours of computation time, and given that there were less computationally expensive models that performed better, we opted not to cross-validate these. We suspect that we have not lost a significant amount of information by not including these two models, as the results as seen in table 2 show that the results from cross-validation largely mirror those of the single train/test split. Note that the metrics are given as the mean values across the three folds.

Model	Mean RMSE	Mean MAE	Mean MAPE	Mean Accuracy
Linear Regression	1834.69	1522.86	117.37%	6.45%
Random Forest	1259.23	969.28	64.93%	83.39%
KNN Regression	888.78	600.48	31.71%	86.54%

Table 2: Performance Metrics of Time-Series Cross-Validation

We again see that all models are improvements over the base linear model, but KNN regression stands out as the best performing. We can therefore be more confident about the robustness of our results—that is that KNN regression is the most ideal method to forecast taxi demand for our needs.

Below, we have included graphs showing the predicted vs. actual values for different models in the single train/test split.

Baseline Linear Regression

Predictors: Pickup hour, Day, Month, Day of Week

For additional linear regression models, please see appendix A

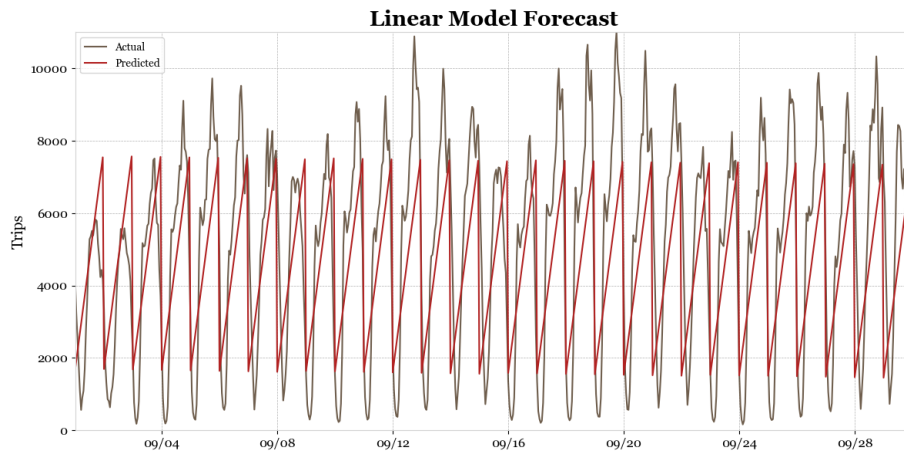


Figure 3: Baseline Linear Regression Forecast

Model Statistics

Metric	Value
RMSE	2007.47
MAE	1716.09
MAPE	93.55%
Accuracy	6.45%
R^2	0.49

SARIMA(2, 0, 3)x(2, 1, 0)₂₄

Exogenous Variables: Pickup hour, day, month, day of week, holiday indicator

For additional SARIMA models, please see appendix B

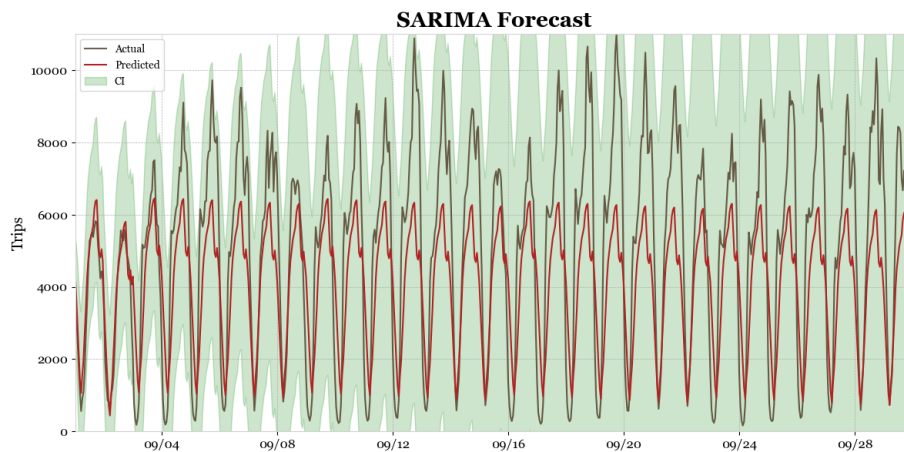


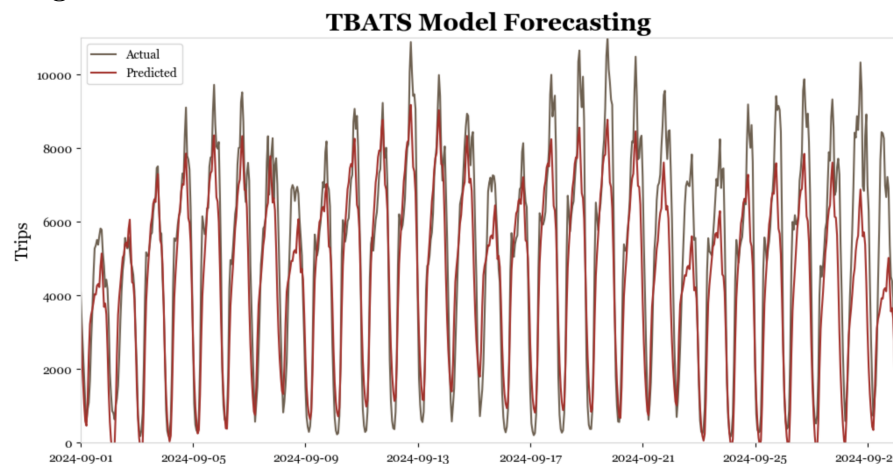
Figure 4: SARIMA with Exogenous Variables Forecast

Model Statistics

Metric	Value
RMSE	2060.26
MAE	1698.39
MAPE	68.74%
Accuracy	31.26%

TBATS

Exogenous Variables: None



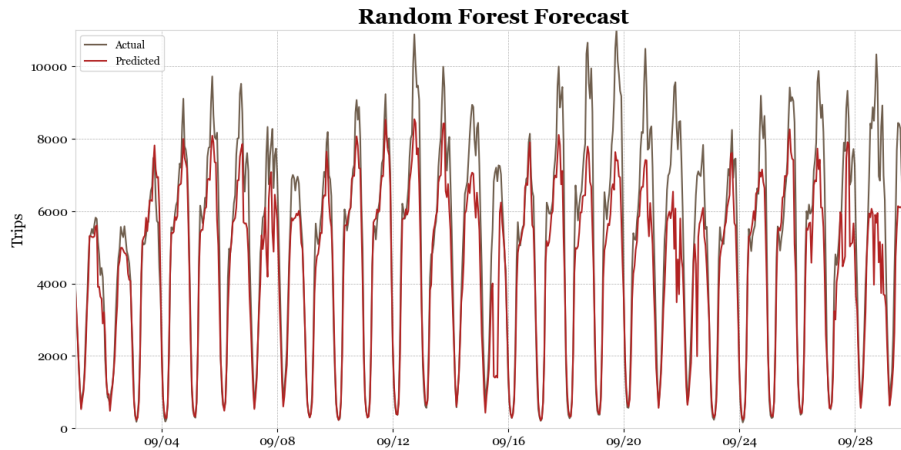
Model Statistics

Metric	Value
RMSE	1370.88
MAE	1056.94
MAPE	36.35%
Accuracy	63.65%

Figure 5: TBATS Forecast

Random Forest

Predictors: Pickup hour, day, month, day of week, holiday indicator, weather
For additional Random Forest models, please see appendix C



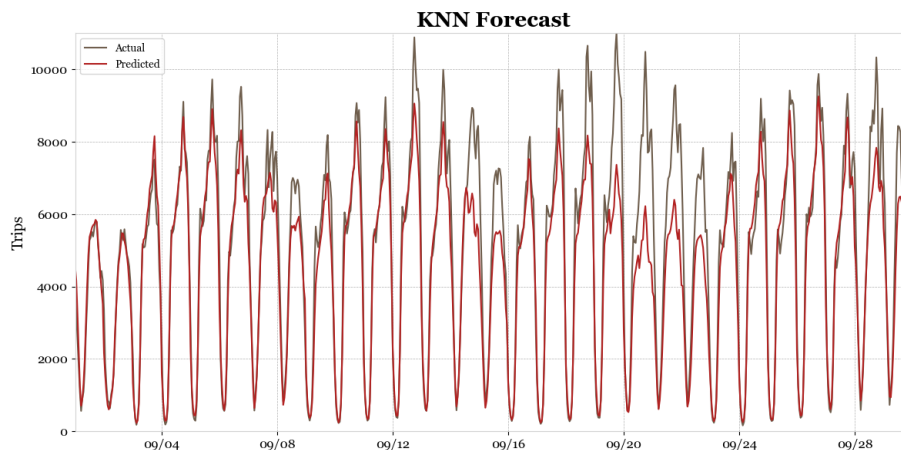
Model Statistics

Metric	Value
RMSE	1251.71
MAE	807.56
MAPE	14.78%
Accuracy	85.22%

Figure 6: Random Forest with Weather Forecast

KNN

Predictors: Pickup hour, day, month, day of week, holiday indicator, Fourier terms
For additional KNN models, please see appendix D



Model Statistics

Metric	Value
RMSE	1041.37
MAE	689.07
MAPE	13.46%
Accuracy	86.54%

Figure 7: KNN with Fourier Terms Forecast

We can see that visually, the predictions appear to match the actual values quite well for the majority of the models. The models that can account for multiple elements of seasonality like RF, TBATS, and KNN are predicting the troughs and peaks particularly well. This is very valuable within the context of our project because at the end of the day, we are trying to especially predict spikes/drops in taxi demand. The relative increases/decreases are arguably more important than the exact demand numbers themselves because city planners, drivers, commuters, etc. really need to know peak/low demand times.

Conclusion

We believe that our experiments effectively show that hourly NYC yellow taxi demand can be reasonably forecasted using the correct models and inputs. We created a number of models that were an improvement over the linear baseline, but K-Nearest Neighbors (KNN) regression with input variables including the hour, day of week, month, day of month, holiday information, and Fourier terms stood out as the most effective approach.

Throughout this process, we have learned valuable lessons about the different advantages/disadvantages of each forecasting model. It was particularly challenging to work with data that had multiple elements of seasonality because we had to research and implement methods that were not covered in the course. The biggest limitation of our project is that we have not predicted by location. New York City is a very large urban area, and it would be particularly useful to forecast demand by zone. Certain areas are definitely more busy than others depending on

the time of day, day of week, etc. A great future extension of our ideas would be to implement this work and to study whether different models need to be developed for different areas of the city. It would also be interesting to forecast green taxis and even Uber/Lyft demand as well to give a more holistic view of ride-hailing patterns across the city as a whole.

We are thankful to have worked on such an interesting project and are grateful to the instructor and TA's of the course for their guidance throughout the process.

References

- [1] NYC Taxi and Limousine Commission (TLC) Trip Record Data
- [2] Z. Li, J. Liu, and X. Yang, "New York city taxi demand forecast based on ARIMA model," *Appl. Comput. Eng.*, vol. 68, pp. 143-149, 2024. [Online]. Available: <https://www.ewadirect.com/proceedings/ace/article/view/13070>. [Accessed: Dec. 10, 2024].
- [3] C. Yang and E. J. Gonzales, "Modeling taxi demand and supply in New York City using large-scale taxi GPS data," in *Seeing Cities Through Big Data*, P. Thakuriah, N. Tilahun, and M. Zellner, Eds. Cham, Switzerland: Springer, 2017, pp. xx-xx. [Online]. Available: https://doi.org/10.1007/978-3-319-40902-3_22. [Accessed: Dec. 10, 2024].
- [4] P. S. Benarji, P. S. Bharadwaj, B. Neeha, D. Srikanth, and V. Ankitha, "Taxi demand prediction using machine learning," *Int. Res. J. Eng. Technol. (IRJET)*, vol. 10, no. 5, pp. 1527-1531, May 2023. [Online]. Available: <https://www.irjet.net/archives/V10/i5/IRJET-V10I5234.pdf>. [Accessed: Dec. 10, 2024].
- [5] X. Li, Y. Zhang, and Z. Chen, "Taxi demand and fare prediction with hybrid models: enhancing efficiency and user experience in city transportation," *Electronics*, vol. 13, no. 18, p. 10192, Sep. 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/13/18/10192>. [Accessed: Dec. 10, 2024].
- [6] T. Cunha, E. C. e Silva, and F. Ferreira, "Individual water consumption forecasting: evaluating SARIMA and TBATS methods," in *2024 7th International Conference on Mathematics and Statistics (ICoMS 2024)*, Amarante, Portugal, Jun. 23-25, 2024, New York, NY, USA: ACM, pp. 1-6. [Online]. Available: <https://doi.org/10.1145/3686592.3686602>. [Accessed: Dec. 10, 2024].
- [7] Y. Zhang, Z. Liu, and X. Zhang, "Short-term demand forecasting of urban online car-hailing based on the k-nearest neighbor model," *Sensors*, vol. 22, no. 23, p. 9456, Nov. 2022. [Online]. Available: <https://www.mdpi.com/1424-8220/22/23/9456>. [Accessed: Dec. 10, 2024].
- [8] S. Ahmed, M. S. Hossain, and A. E. Hassan, "Applying k-nearest neighbors to time series forecasting: two new approaches," *arXiv preprint arXiv:2103.14200*, Mar. 2021. [Online]. Available: <https://arxiv.org/abs/2103.14200>. [Accessed: Dec. 10, 2024].
- [9] Open-Meteo Weather API

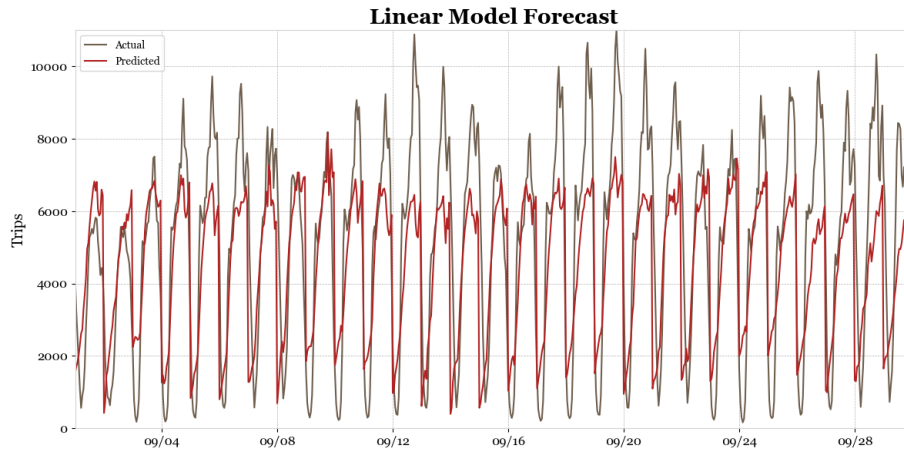
Appendix

This appendix contains visuals of models we created but decided not to use in our final selection.

A: Linear Regression

Linear Regression with Holiday Indicator and Weather

Predictors: Pickup hour, day, month, day of week, holiday indicator, weather



Model Statistics

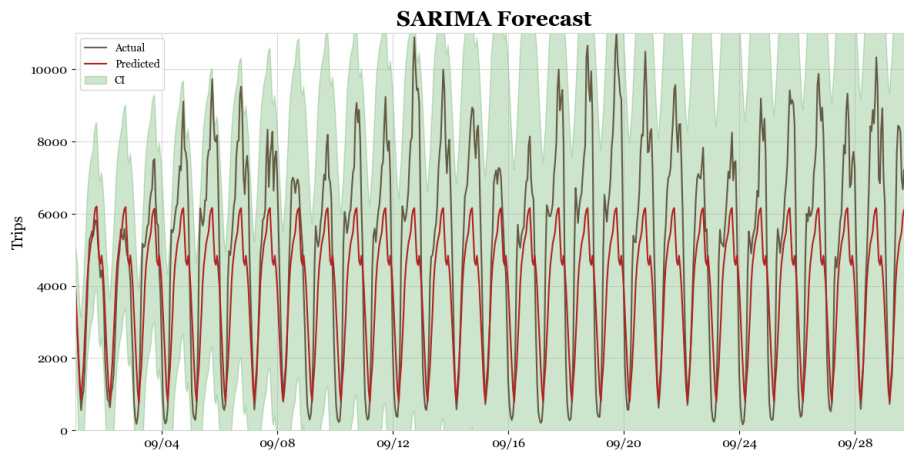
Metric	Value
RMSE	1850.86
MAE	1449.48
MAPE	76.21%
Accuracy	23.79%
R^2	0.56

Figure 8: Linear Regression with Holiday Indicator and Weather Forecast

B: SARIMA

SARIMA(2, 0, 3)x(2, 1, 0)₂₄ with no Exogenous Variables

Predictors: None



Model Statistics

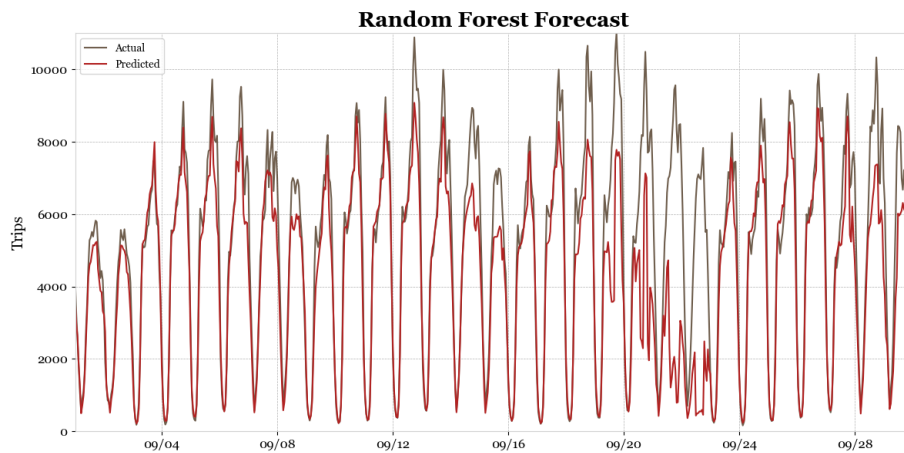
Metric	Value
RMSE	2060.26
MAE	1698.39
MAPE	68.74%
Accuracy	31.26%

Figure 9: SARIMA Baseline Forecast

C: Random Forest

Random Forest Baseline

Predictors: Pickup hour, day, month, day of week



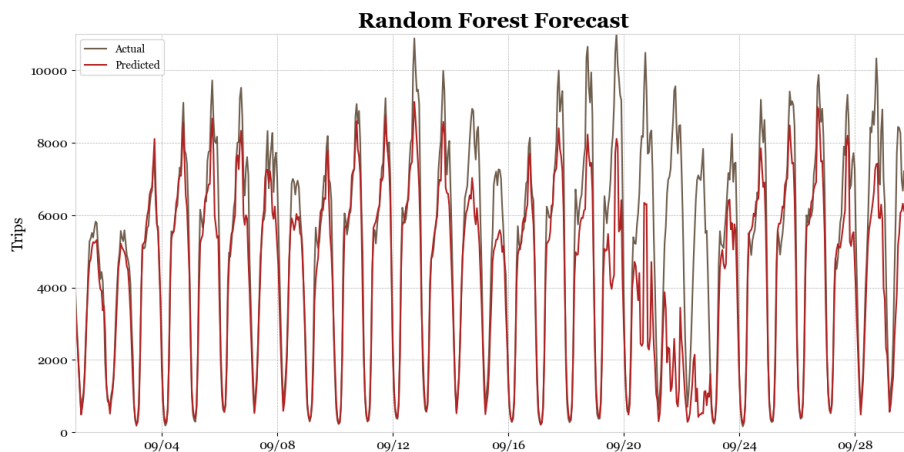
Model Statistics

Metric	Value
RMSE	1570.11
MAE	902.09
MAPE	16.49%
Accuracy	83.51%

Figure 10: Random Forest Baseline Forecast

Random Forest with Fourier Terms

Predictors: Pickup hour, day, month, day of week, indicators for holiday, day before holiday, day after holiday, Fourier transformations of hour, day of week, and month



Model Statistics

Metric	Value
RMSE	1627.89
MAE	946.30
MAPE	17.53%
Accuracy	82.47%

Figure 11: Random Forest with Fourier Terms Forecast

Random Forest with Weather and Fourier Terms

Predictors: Pickup hour, day, month, day of week, weather, indicators for holiday, Fourier transformations of hour, day of week, and month

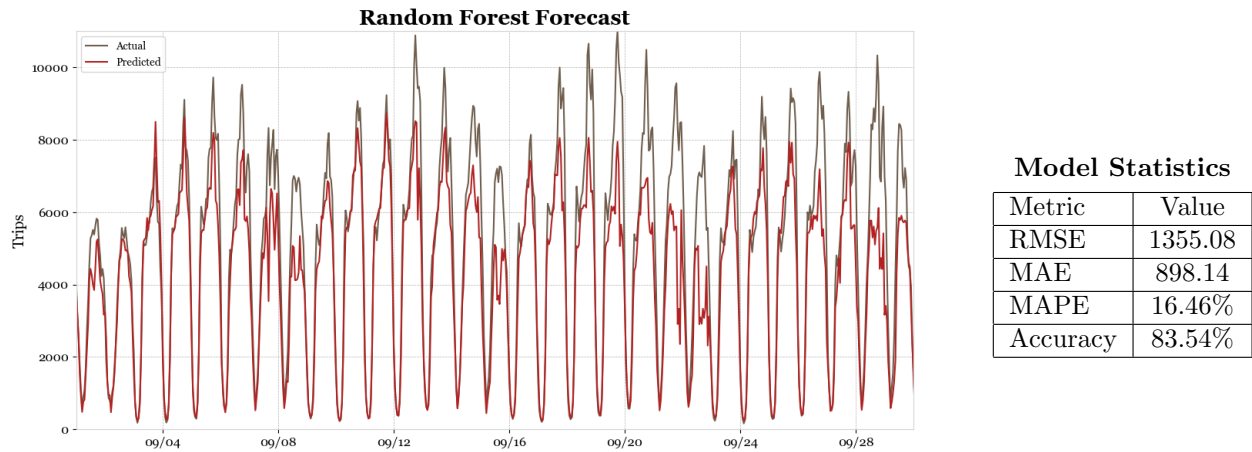
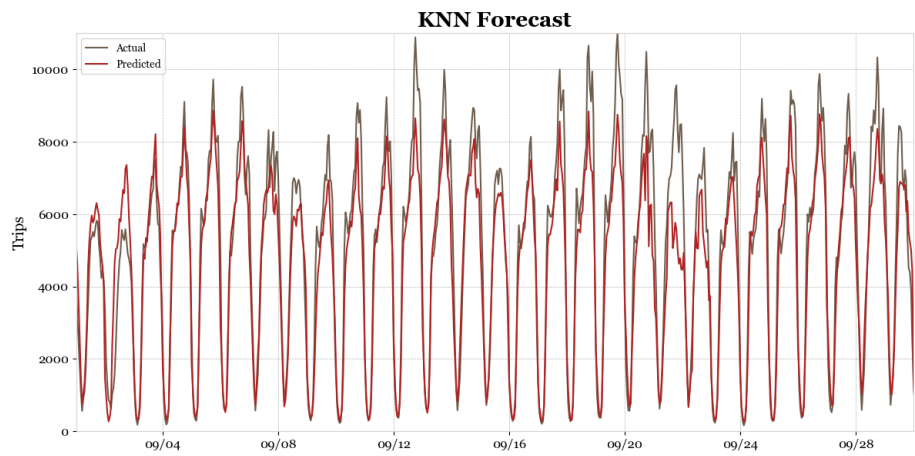


Figure 12: Random Forest with Weather and Fourier Terms Forecast

D: KNN

KNN Baseline

Predictors: Pickup hour, day, month, day of week



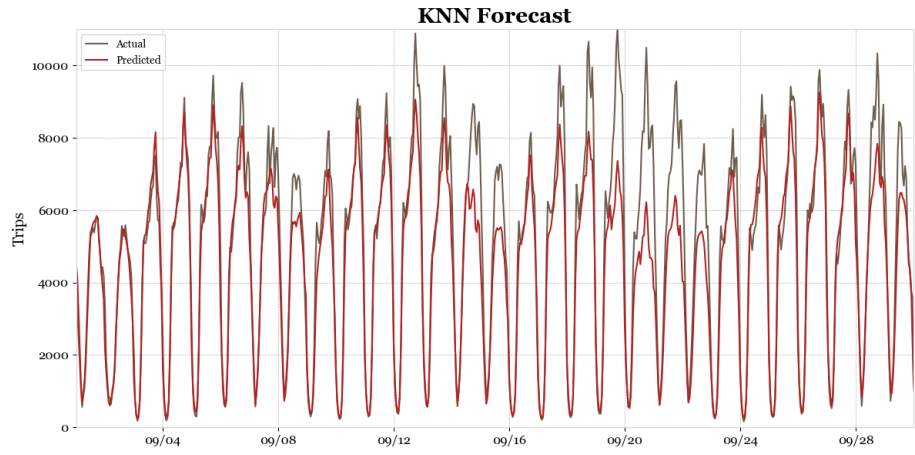
Model Statistics

Metric	Value
RMSE	1005.55
MAE	714.95
MAPE	16.90%
Accuracy	83.10%

Figure 13: KNN Baseline Forecast

KNN with Weather

Predictors: Pickup hour, day, month, day of week, holiday indicator, weather



Model Statistics

Metric	Value
RMSE	1746.28
MAE	1330.36
MAPE	52.77%
Accuracy	47.23%

Figure 14: KNN with Weather Forecast